

4 (a) State the value of absolute zero on:

(i) the Celsius temperature scale

temperature = ..... °C [1]

(ii) the thermodynamic temperature scale. Give a unit with your answer.

temperature = ..... unit ..... [1]

(b) A sample contains a fixed amount of gas. The gas has pressure  $p$ , volume  $V$  and thermodynamic temperature  $T$ .

Fig. 4.1 shows the variation of  $pV$  with  $kT$  for the sample, where  $k$  is the Boltzmann constant.

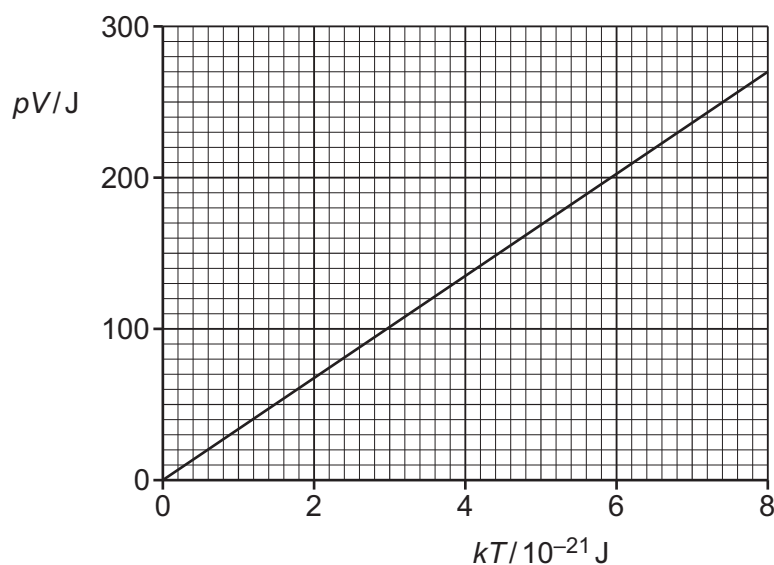


Fig. 4.1

(i) State what is indicated about the nature of the gas from the variation shown in Fig. 4.1.

..... [1]

(ii) Determine the number  $N$  of molecules of the gas in the sample.

$N =$  ..... [2]

